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A PALEOMAGNETIC STUDY OF CONCENTRATING IRON ORES FROM NORTHERN MICHIGAN

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ABSTRACT

Fifty samples were collected from the concentrating iron ores and associated rocks in northern Michigan, U.S.A. They came from: the Republic Mine, Marquette Range; the Groveland Mine, Felch Mountain District; and along the Menominee Range. Standard paleomagnetic procedures were used to study the samples. The Negaunee and Vulcan iron formations are thought to have been deposited as primary sediments of the hematite-banded and magnetite-banded oxide facies with chemical remanent magnetization (CRM) during the Middle Animikie. During dynamic or thermal metamorphism, the primary remanence directions were modified causing them to be rotated into the bedding plane to yield directions that are random in declination but which are uniformly distributed about the bedding plane in inclination. The metamorphism was likely associated with the Penokean orogeny during which the iron formations were tightly folded and then thermally metamorphosed. Paleomagnetism suggests that the iron formation at the Republic and Groveland Mines has been folded into synclines. The Lower Keweenawan diorites have a post-folding thermal remanent magnetization (TRM). The overlying Cambrian sandstones have a primary CRM or detrital remanent magnetization (DRM). The above paleomagnetic conclusions are consistent with the geologic evidence. The paleomagnetic pole position determined from the Keweenawan diorites and the Cambrian sandstones are thought to be fairly reliable.

INTRODUCTION

In a previous study (31), the author found that unmetamorphosed iron formation from the Gunflint, Mesabi and Cuyuna iron ranges has a stable primary remanence. The purpose of this study was to see if metamorphosed iron formation from three localities in northern Michigan (Fig. 1) has a similar remanence and if paleomagnetism would prove useful in studying the genesis of the concentrating iron deposits in the iron formation. The three iron formations are thought to be contemporaneous; they have been subjected to low-, intermediate-, and high-grade thermal metamorphism, and to moderate and intense dynamic metamorphism; and they are mined as concentrating iron ores at two of the localities. Most of the samples for this study come from the iron formations and a few samples come from the associated Keweenawan diorites and Cambrian clastic sediments. Twenty samples come from the Republic Mine on the Marquette iron range, fourteen samples from along the Menominee iron range, and sixteen samples from the Groveland Mine in the Felch Mountain iron district. The samples from these three localities are identified by the prefix letters, "B," "C," and "D" respectively before the sample number. In this paper the results from each of

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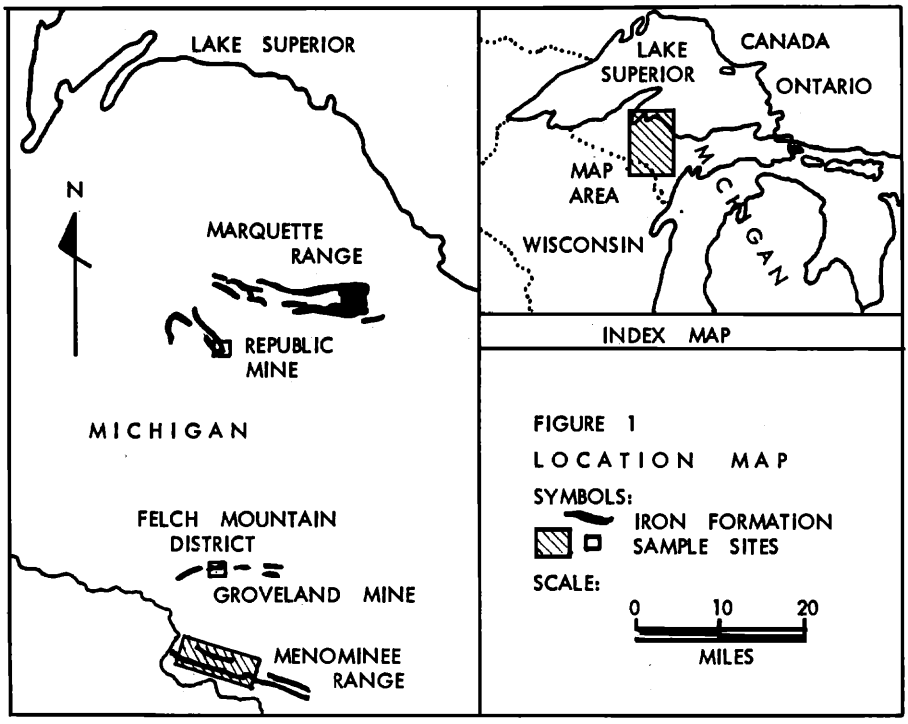


FIG. 1. Location map of northern Michigan, U.S.A.

these three localities will be presented in turn and discussed in the conclusion.

The samples were oriented in relation to features on surveyed maps, and the orientation is estimated to be accurate to within 5° of arc at the 95% confidence level. The magnetic measurements were made on an astatic magnetometer with a sensitivity of 5×10^{-8} emu/gm using a system of measurement similar to that described by As (1). Alternating field (AC) demagnetization was done using fairly standard equipment. The statistics follow the equations of Fisher (9). Details on the paleomagnetic theory, equipment and methods used in this study and the individual measurements are given elsewhere (30, 31).

REPUBLIC MINE

Geology.—Van Hise and Leith (34) have summarized the classic geologic study of the Marquette Range by Van Hise and Bayley (33) at the turn of the century. The stratigraphic sequence shown in Table 1 is taken from a recent geologic summary by Boyum (4). This sequence has been fitted into the subdivision of Precambrian time which Goldich *et al.* (11) proposed.

At the Republic Mine, shown in Figure 2, the Negaunee iron formation of Middle Animikie age is 1400 feet thick (4). Goldich *et al.* (11) report two K-Ar radioactive age determinations of 1.61 and 1.89 billion (10^9)

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years for samples from the Negaunee iron formation which indicate its minimum age and correspond to the age of the Penokean orogeny. The uppermost iron formation belongs to the hematite-banded oxide facies, and it overlies unsampled iron formation of the nongranular silicate facies (18). The iron formation has been metamorphosed to at least the sillimanite isograd of thermal metamorphic grades (19). The hematite-banded oxide facies is mined as an iron ore of concentrating grade, which is called "jaspilite ore". The jaspilite ore occurs with either iron formation of the nongranular silicate facies or Keweenawan diorite as its footwall and with the Goodrich quartzite as its hanging wall.

The Goodrich quartzite is composed of quartzite with interbeds of slate and conglomerate. It is the lowermost formation of the Michigamme series of clastic and pyroclastic sediments and it overlies the Negaunee iron formation with an angular unconformity of up to 15°. The Michigamme series is in excess of 3000 feet thick. It represents the Upper Animikie system.

The Keweenawan diorite intrusives occur as sills and dikes in the Negaunee iron formation. There are three prominent sills with an aggregate thickness of about 1400 feet and they appear conformable to the bedding of the iron formation. One sill occurs on the east side of the Republic Mine and a diorite dike is found on the south side as shown in Figure 2. Recently

TABLE 1

STRATIGRAPHIC SECTIONS FOR THE SAMPLING SITES

ERA	AGE YEARS $x(10^9)$	PERIOD (Orogeny)	SYSTEM	MARQUETTE IRON RANGE	MENOMINEE IRON RANGE	FELCH MOUNTAIN IRON DISTRICT
PALEOZOIC	0.6	CAMBRIAN			sandstone	sandstone basal conglomerate
	1.0	(Grenville)			angular unconformity	angular unconformity
LATE PRECAMBRIAN		KEWEENAWAN	LOWER(?)	dioritic intrusives	diabase intrusives	diabase intrusives
	1.7	(Penokean)		intrusive contacts	intrusive contacts	intrusive contacts granitic intrusives intrusive contacts metadiabase and metagabbro intrusives intrusive contacts
MIDDLE PRECAMBRIAN		ANIMIKIE	UPPER	Michigamme series Goodrich quartzite	Michigamme slates	Badwater greenstone Michigamme slates Hemlock felsite formation
				angular unconformity	unconformity	unconformity
			MIDDLE	Negaunee iron formation Siamo slate Ajibik quartzite	Vulcan iron formation Felch shist	Vulcan iron formation Felch shist and quartzite
				angular unconformity	paraconformity	paraconformity
			LOWER	clastic sediments	dolomite, quartzite and conglomerate sediment	dolomite, quartzite and conglomerate sediment
	2.5	(Algoman)		profound unconformity	profound unconformity	profound unconformity
EARLY PRECAMBRIAN				granite, gneiss and greenstones	granite, gneiss and shist	granite, gneiss and shist

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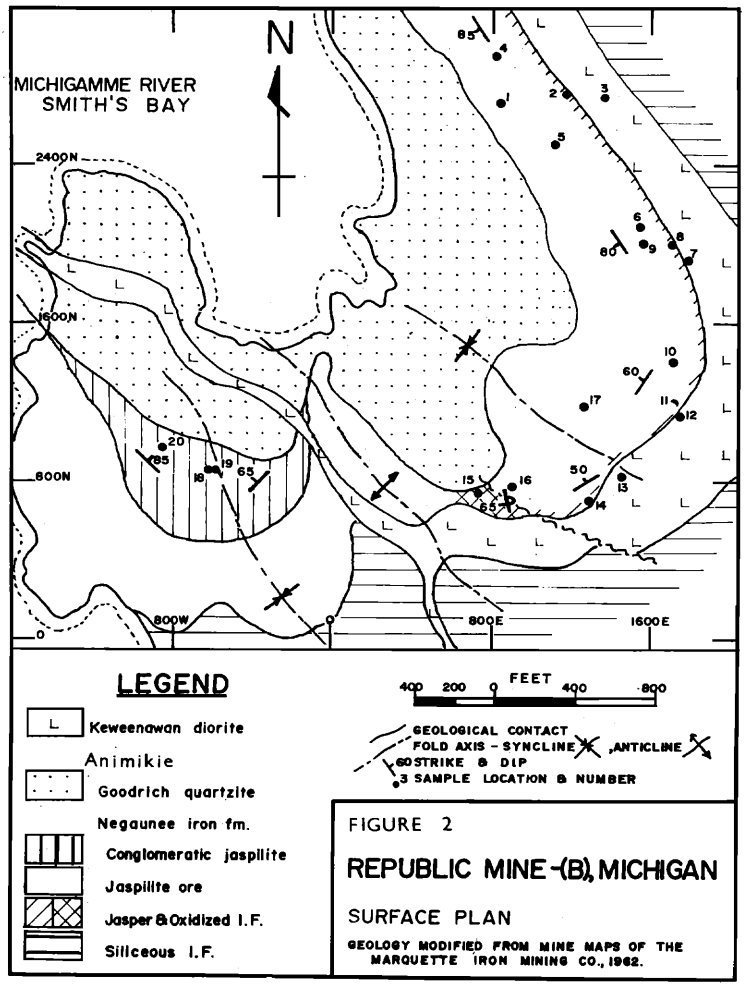


FIG. 2. Map of the Republic Mine, Marquette iron range.

Symons (32) has provided paleomagnetic evidence to show that these sills on the Marquette Range are post-folding intrusives rather than conformable extrusives as some geologists have proposed (34).

Origin of the Ore.—James (18) believes that the jaspilite ore was deposited as iron formation of the hematite-banded oxide facies in an oxidizing environment at the end of Middle Animikie. Subsequent high-grade metamorphism created the specular hematite.

Royce (26) argues that surface waters oxidized the primary siderite in the iron formation to hematite at the end of the Middle Animikie. Subsequent metamorphism converted the oxidized iron formation into jaspilite ore.

Van Hise and Leith (with Mead) (34) state that meteoric solutions of

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Upper Animikie age flowed down along a trough formed by the folded extrusive "diorite" sheet. These solutions oxidized the siderite in the primary iron formation to hematite. Later metamorphic and intrusive activity converted the oxidized iron formation into jaspilite ore.

Gruner (15, 16) suggests that hydrothermal solutions, derived from an igneous source or from groundwater heated by an igneous source, circulated in the Negaunee iron formation between the impermeable diorite sill and Goodrich quartzite. The solutions oxidized the primary iron minerals in the iron formation to specular hematite.

Boyum (4) finds that the jaspilite ores of the Marquette Range are related to structural features such as folds and to diorite intrusives. Also, the ores show replacement features and a higher metamorphic grade than the containing iron formation. Therefore, he appears to support an epigenetic

Symons (32) on paleomagnetic evidence concludes that the stratigraphically equivalent and structurally similar hard hematite ore deposit of the Cliff-Shaft Mine, Marquette Range, is syngenetic. He believes that it was formed as a laterite on exposed iron formation of the magnetite-banded oxide facies at the end of the Middle Animikie.

Paleomagnetic Results.—At the Republic Mine, seventeen samples (B-1 to 17) were collected from within the mine open pit around the main synclinal fold, and three samples (B-18 to 20) from a drift-stripped subcrop around a subsidiary synclinal fold as shown in Figure 2. The magnetic characteristics of the samples are summarized in Table 2.

Nine samples (B-1, 4, 5, 6, 9, 10, 14, 16 and 17) of jaspilite ore were collected. The ore is composed of fine bands of specular hematite and chert with some magnetite-rich and red hematite-rich laminations. The prominent preferred orientation in the distribution of the specular hematite is shown in Figure 3a. The jaspilite ore yields a stable remanence after AC (alternating field) demagnetization as shown in Figure 4a where the remanence intensity is unchanged between the 300 and 700 oersted (peak) steps. This unchanged remanence and the range of magnetic characteristics of the samples (Table 2) are consistent with the hypothesis that their remanence originated as a chemical remanent magnetization (CRM). Unfolding the synclinal structure causes the precision parameter to decrease from 1.3 to 1.2. Both values are so low that no statistical significance can be attached to either the prefolding or postfolding grouping of remanence directions. However, their mean inclination is only $3.5° \pm 15°$ (s.d.) from the bedding plane and their declinations are randomly oriented as shown in Figure 5b which indicates a strong bias of the remanence direction to the bedding plane of the jaspilite ore. These jaspilite ore samples exhibit a moderate magnetic anisotropy as shown by the susceptibility differences measured along orthogonal axes. The measurements show that on average, the maximum susceptibility is at least 25 percent greater than the minimum susceptibility and approximately parallel to the bedding plane. The bias of the remanence direction to the bedding plane and the apparent anisotropy in the bedding plane suggest that the primary remanence direction has been modified and rotated into the bedding plane by metamorphism.

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TABLE 2

SUMMARY OF THE MAGNETIC MEASUREMENTS

SAMPLE GROUP	N	X	Ms	Mn/Ms	MEAN REMANENCE VECTOR					POLE POSITION			
					R	Dm	Im	$\alpha 63$	k	Dp	Ip	dp	dm
REPUBLIC MINE*													
Jasper iron formation	2U	556-1551	14-154	5-6	2.0	226	86	10	52	99	45	21	21
Oxidized iron formation	1C	34	18	2		222	39						
Conglomeratic iron formation	6	65-187	12-84	2-15									
Jaspilite ore	9C	10-281	4.6-25	1.5-15	2.7	212	7	46	1.3				
Keweenawan diorite	4U	6.1-363	0.07-5.2	13-140	3.6	248	72	16	8.1	122	28	26	29
MENOMINEE RANGE*													
Vulcan iron formation	13	1.4-306	0.4-21	1-200									
GROVELAND MINE*													
Vulcan iron formation,													
Basal and Lower members	4C	3-537	1.4-8.3	4-23	3.7	74	41	14	10.9	4	27	10	17
Middle and Upper members	8	1-3700	1.4-43	2-80									
Cambrian sandstone**	3C	0.07-4.8	0.07-0.80	1-4	2.7	131	28	23	5.9	39	-15	14	26

* Geographic co-ordinates: Republic Mine, 87.9°W, 46.5°N; Menominee iron range, 88.0°W, 45.8°N; and the Groveland Mine, 87.8°W, 46.0°N.

** Includes one sample from the Menominee iron range.

The symbols and units are:

N—the number of samples represented in the group or the population of the directions,
C (U)—indicates that the remanence directions have been corrected (uncorrected) for tectonic deformation prior to calculation of the mean remanence direction and pole position,
X—the range of susceptibilities in (emu/gm $\times 10^{-6}$),
Ms—the range of intensities of the stable remanences in (emu/gm $\times 10^{-6}$) after AC demagnetization at 700 oersteds (peak),
Mn/Ms—the range of ratios of NRM intensity (Mn) to Ms intensity which roughly indicates the ratios of unstable (Mn-Ms) to stable (Ms) remanence,
R—the length of the mean remanence vector with each of the N remanence directions considered as a unit vector,
Dm—the declination of the mean remanence vector in degrees reckoned clockwise from 0°,
Im—the inclination of the mean remanence vector in degrees reckoned from the horizontal plane and considered positive (negative) if directed downwards (upwards),
$\alpha 63$—the semi-angle of the circular standard deviation, in degrees, of the cluster of remanence directions,
k—the precision parameter of the cluster of remanence directions,
Dp (Ip)—the longitude (latitude) of the virtual paleomagnetic pole position in degrees,
dp (dm)—the semi-axis of the oval of confidence derived from the circular standard deviation perpendicular to (along) the meridian between the sampling site and the pole position in degrees about the pole position.

Three samples of jasper iron formation were collected from the Negaunee iron formation from within 50 feet of its contact with the diorite sill (Fig. 2). One sample (B-8) was excluded from further consideration because it had a very large unstable remanence moment of about one hundred times the intensity of the moment after AC demagnetization which suggests magnetic instability. The remaining two samples (B-2 and 11) are composed of banded chert with some magnetite laminations. Unfolding causes the precision parameter for their remanence directions to decrease slightly from

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52.2 to 45.4, which indicates that the fold test is not significant or that the remanence may have been acquired either before or after folding. The remanence directions before tectonic correction are in very close agreement with those of the Keweenawan diorite (Figs. 5a, 5d). This agreement suggests that the jasper remanence is a TRM acquired by contact metamorphism at the time of diorite intrusion. The strong remanent and induced magnetic intensities of the jasper samples (Table 2) also suggest a thermal remanent magnetization (TRM).

One sample (B-15) of oxidized iron formation was collected from a minor overturned fold next to a minor fault (Fig. 2). The sample's magnetic characteristics are very similar to those of the jaspilite ore (Table 2, Fig. 5a).

Three field samples of conglomeratic iron formation were collected from

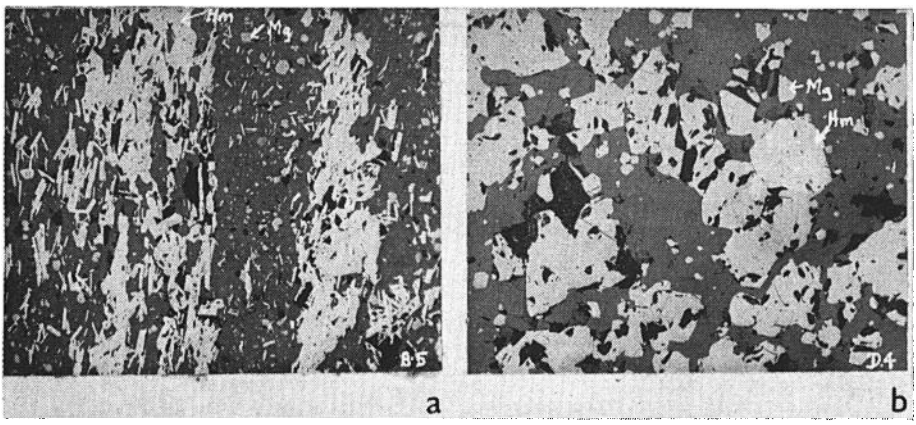


FIG. 3. Microphotographs showing: (a) Jaspilite ore from the Republic Mine (B-5). The euhedral laths of specular hematite are oriented anisotropically or parallel to the bedding laminations. There are a few grains of euhedral magnetite co-existing with the hematite. × 70. (b) Magnetite-banded oxide facies of Vulcan iron formation from the Groveland Mine (D-4). The magnetite is euhedral and coarse grained, and it is co-existing with minor amounts of euhedral hematite. × 70.

which six laboratory samples (B-18A, 18B, 19A, 19B, 20A and 20B) were measured. Their magnetic characteristics are very similar to those of the jaspilite ore (Table 2). Samples B-18A and B18-B have similar remanence directions as do samples B-20A and B-20B (Fig. 5c). However, samples B-19A and B-19B yield divergent remanence directions. The principal magnetic mineral is hematite, which gives a stable remanence after AC demagnetization (Fig. 4b) and this does not change in direction by more than 10° of arc between the 100 and 700 oersted steps. Fegan (10) believes that the conglomeratic iron formation was deposited by sedimentary processes on the pre-upper Animikie erosion surface in which case the divergence between samples B-19A and B-19B indicates that the remanence is stable and has been preserved since deposition. Gross (14) believes that the conglomeratic iron formation could be brecciated iron formation that was formed during folding

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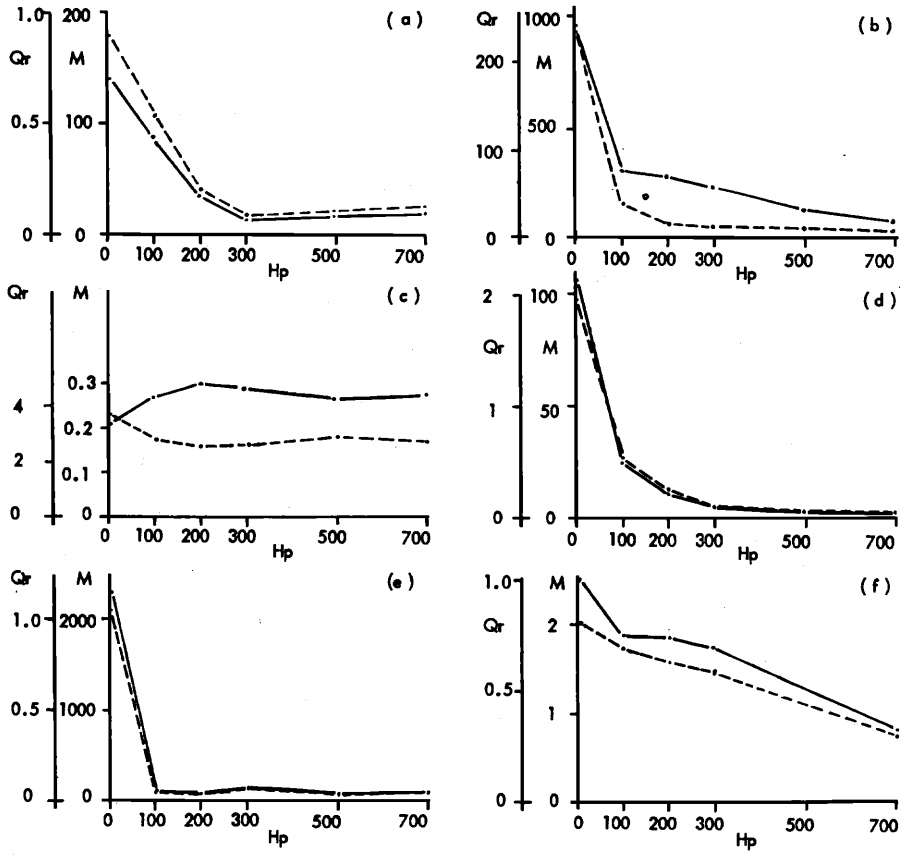


FIG. 4. AC demagnetizing curves for samples from the: (a) Jaspilite ore, Republic Mine (B-5). Its mineralogy is described in Figure 3a. (b) Conglomeratic iron formation, Republic Mine (B-20A). Around the irregularly-shaped blebs of gangue, there are coarse euhedral crystals of hematite which occur in poorly defined bands. (c) Cambrian sandstone, Menominee Range (C-6). The rounded quartz grains or siliceous oölites are surrounded by hematitic "dust" in a siliceous cement. (d) Vulcan iron formation, Menominee Range (C-11). The fine-grained crystals of magnetite have been corroded and heavily altered to hematite. The magnetite and hematite occur in bands. (e) Upper iron-bearing member, Vulcan iron formation, Groveland Mine (D-4), as described in Figure 3b. (f) Cambrian conglomerate, Groveland Mine (D-9B). A hematite pebble in the conglomerate which consists of an irregular patchwork of microcrystalline hematite. The symbols on the graphs are: M, the intensity of the remanence in (emu $\times 10^{-5}$/gm) which is traced by the solid line; Qr, the ratio of the remanence intensity to the unit-field induced intensity which is traced by the dashed line; and Hp, the intensity of the AC demagnetizing field in oersteds (peak).

such as that found at Fort Gouraud in Mauritania (13) in which case the divergence indicates remanence stability since folding.

Four samples (B-3, 7, 12 and 13) were collected from the diorite sill. They grade from fine-grained diorite to coarse-grained amphibolite. The

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samples have fairly high unstable to stable remanence intensity ratios and low Koenigsberger ratios (i.e., this is the ratio of the intensity of the remanent moment to the intensity of the moment induced by a unit field) (Table 2) which suggest poor remanence stability. However, before tectonic correction, the remanence directions form a group with a precision parameter

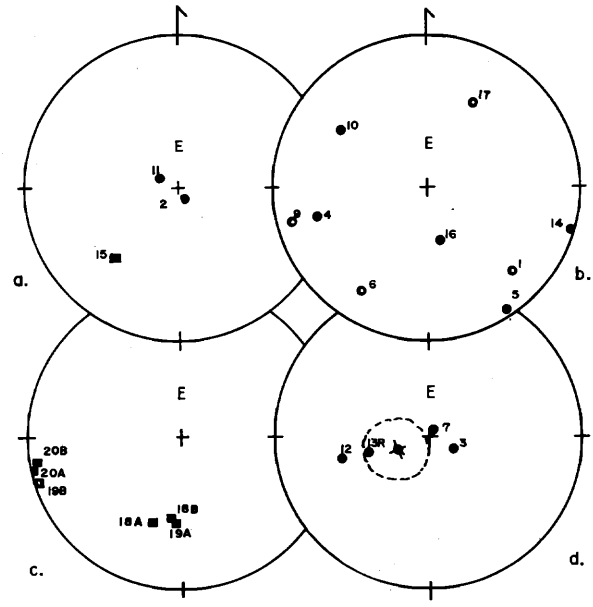


FIG. 5. Remanence directions for samples from the Republic Mine: (a) Jasper iron formation (circles) and oxidized iron formation (square) of the Negaunee formation before and after applying tectonic corrections respectively, (b) Jaspilite ore after applying tectonic corrections, (c) Conglomeratic iron formation after applying tectonic corrections, and (d) Keweenawan diorite before applying tectonic corrections. The corrections were made by rotating the strike of the sample to S40°E and the dip to the horizontal. The directions are plotted on Schmidt equal-area polar projections with declinations reckoned clockwise from the north position and inclinations reckoned from the perimeter. Positive (negative) or downwardly (upwardly) directed inclinations are indicated by solid (open) symbols. The mean direction of a group of samples is indicated by a cross over the group symbol with its cone of circular standard deviation indicated by a dashed line. The direction of the Earth's present magnetic field at the sampling site is indicated by the symbol E.

of 8.1 and have a mean remanence direction (Fig. 5d) similar to Lower Keweenawan directions determined by others (*7*, 12, *29*, 32) for the Lake Superior area. The mean remanence direction of the diorite is in close agreement with the jasper iron formation, but it is unrelated to the directions of the jaspilite ore, conglomeratic iron formation and oxidized iron formation (Fig. 5) which must have been formed prior to, or during, intrusion of the diorite.

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MENOMINEE RANGE

Geology.—At the turn of the century, W. S. Bayley (3) mapped and reported on the geology of the Menominee iron range. Van Hise and Leith (34) summarized his work. Two useful summaries that incorporate more recent work are by Dutton and Lamey (8) and R. W. Bayley (2). The stratigraphy of the range given in Table 1 is taken from James (20).

The Vulcan iron formation of Middle Animikie age is divided into three members on the Menominee iron range: the basal Traders member contains mostly iron formation of the magnetite-banded oxide facies; the middle Brier member contains slate with a minor graphite, carbonate and magnetite; and the upper Curry member contains mostly iron formation of the hematite-banded oxide facies (18, 20). James (19) considers that the iron formation belongs to the chlorite-biotite metamorphic facies. Goldich *et al.* (11) report one K-Ar radioactive age determination of 1.35 billion (10^9) years for a sample from the underlying sediments of the Chocolay group of Lower Animikie age. This date indicates the minimum age of the Chocolay group's last metamorphic event which is probably the Penokean orogeny.

The Vulcan iron formation of the Menominee range has been regionally folded into a series of anticlinal and synclinal folds with northwesterly striking axes as shown in Figure 6. Superimposed on these regional folds are minor pitching troughs caused by cross-folding or faulting. These troughs in places contained soft hematite (martite) deposits of direct-shipping grade in the Traders member. No deposits of concentrating iron ore have been exploited.

The Cambrian is represented by red sandstones that are flat lying and undisturbed and which unconformably overlie the Precambrian Animikie strata.

Paleomagnetic Results.—Fourteen samples (C-1 to 14) were collected from along the Menominee Range as shown in Figure 6. All the samples were collected from well below the original surface in abandoned mine pits except for one sample (C-1) from a surface outcrop, one sample (C-2) from a road cut, and two samples (C-3 and 11) from the crests of mine pits. Thirteen samples (C-1 to 5 and 7 to 14) were collected from the magnetite-banded and hematite-banded oxide facies of the Vulcan iron formation. Their remanence directions proved to be quite stable after AC demagnetization removed the unstable components as shown by step-demagnetization (Fig. 4d) and reversed-orientation demagnetization tests. The latter tests on seven samples yielded an average change in remanence direction of only 6° of arc. The remanence intensities of the samples are quite weak but fairly uniform (Table 2) whereas their induced intensities are much more intense and varied. Their average anisotropy of magnetic susceptibility is quite prominent with the maximum axis at least 25 percent more intense than the minimum axis. The precision parameter for their remanence directions is 1.9 before unfolding and 1.5 after unfolding, neither of which is statistically significant. However, the inclinations of the remanence directions have an average value of $9° \pm 8°$ (s. d.) and the declinations are quite

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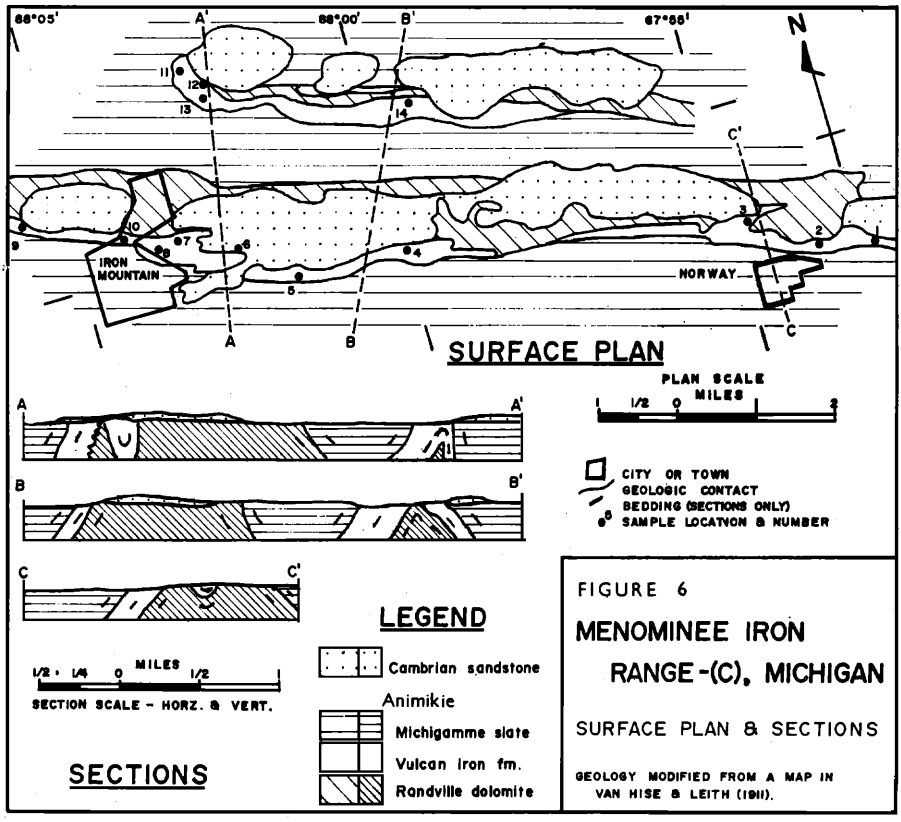


FIG. 6. Map of the Menominee iron range.

random as shown in Figure 7a. This pattern indicates a bias for the remanence directions to be aligned in the bedding plane of the iron formation.

The single sample (C-6) of Cambrian sandstone will be discussed in the next section with the Cambrian samples collected at the Groveland Mine.

GROVELAND MINE

Geology.—The classic geologic study of the Felch Mountain iron district was by Clements and Smyth (5). Subsequent and more detailed studies are reported by James *et al.* (21). Recently Cumberlidge and Stone (6) have given an excellent account of the geology of the Groveland Mine. The stratigraphic sequence given in Table 1 is taken from James *et al.* (21).

At the Groveland Mine, the Vulcan iron formation of Middle Animikie age can be subdivided into four members (6) as shown in Figure 8. The basal quartzite member contains massive vitreous quartzite with minor magnetite and hematite. It varies between 30 and 110 feet in thickness. The lower iron-bearing member belongs to the hematite-banded oxide facies. It

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varies between 10 and 235 feet in thickness. The middle iron-bearing member is transitional and belongs to both the hematite-banded and magnetite-banded oxide facies. It varies between zero and 30 feet in thickness. Finally, the upper iron-bearing member belongs to the magnetite-banded oxide facies. It is over 300 feet thick. The iron formation has been thermally metamorphosed to above the staurolite isograd (19) as indicated by the presence of diopside, Ca-Mg hornblende, cummingtonite, garnet and recrystallized chert (6). Up to 100 feet below the Cambrian-Precambrian unconformity, the iron formation may be oxidized to yield specular hematite, jasper and secondary carbonate. Also, there is a zone of hydrated iron formation associated with intense fracturing, which contains limonite and carbonate, and which occurs immediately below a pyrite zone in the Cambrian sandstone.

The Felch Mountain iron district is an infolded E-W belt of Animikie sediments surrounded by Early Precambrian granites and gneisses. At the Groveland Mine, the three iron-bearing members of the Vulcan iron formation

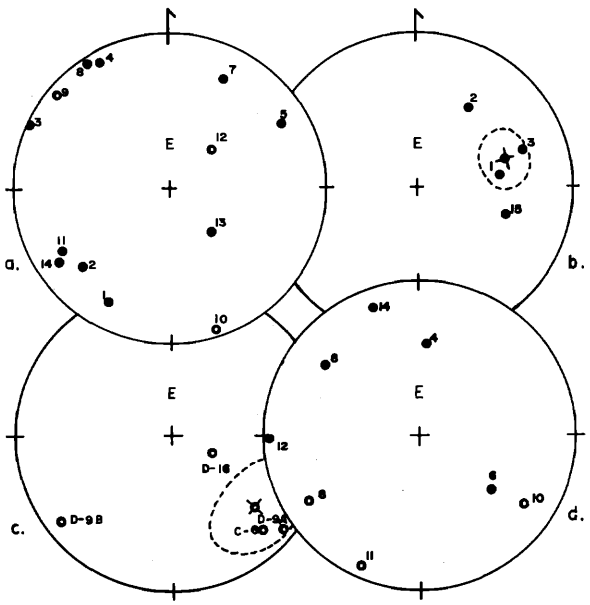


FIG. 7. Remanence directions for samples from: (a) the Vulcan iron formation of the Menominee range after applying tectonic corrections by rotating the strike of the samples to S70°E and the dip to the horizontal, (b) the basal and lower members of the Vulcan iron formation at the Groveland Mine after applying tectonic corrections by rotating the strike to E-W and the dip to the horizontal, (c) the Cambrian sandstone (C-6, D-16) and conglomerate (D-9A and 9B) from the Menominee Range (C) and Groveland Mine (D) after applying tectonic corrections by rotating the dip to the horizontal about the strike (n.b. the cone of circular standard deviation includes samples C-6, D-9A and D-16 only), and (d) the middle and upper members of the Vulcan iron formation at the Groveland Mine after applying tectonic corrections as in (b) above assuming a suncline. The conventions of plotting are th same as for Figure 5.

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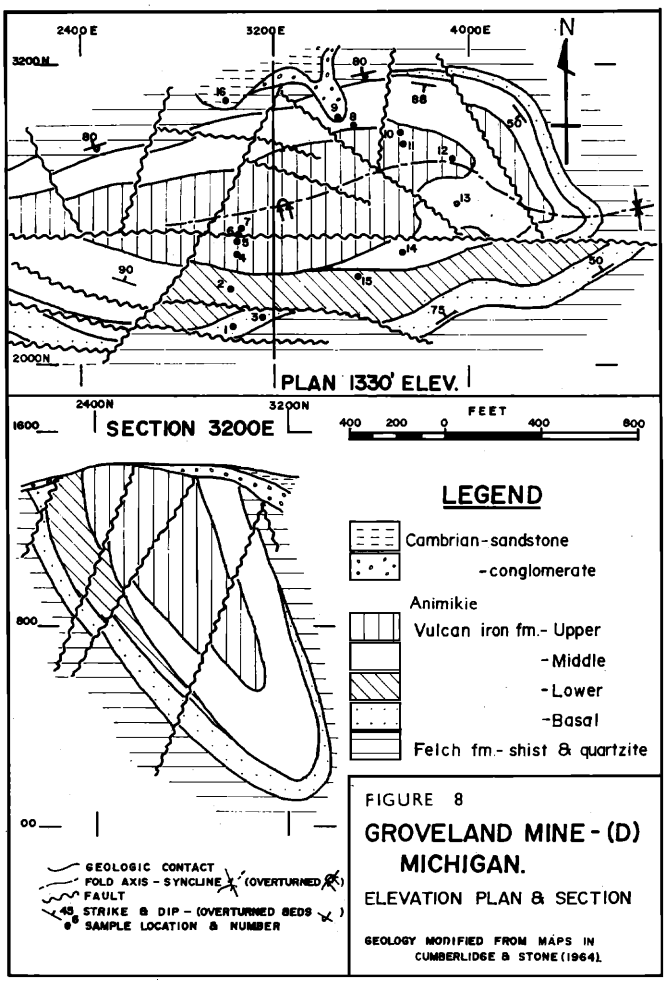


FIG. 8. Map of the Groveland Mine, Felch Mountain iron district.

are mined as an iron ore of concentrating grade. The structure of the deposit has not been clearly established. Cumberlidge and Stone (6) favor the possibility that the structure is a tight overturned syncline with an axial plane that strikes E-W and dips at −60°N as shown in Figure 8. As evidence for the syncline, they cite the rapid thickening of the iron formation at the deposit and the apparent closure of the iron formation around the east end of the deposit. The second possibility is that the structure is a single monoclinal limb for, as Stone (27) states, the synclinal axis has not been found despite the complete rock exposure in the mine pit. There is a major E-W strike-slip fault cutting through the center of the deposit and there are several minor faults as shown in Figure 8. The deposit is cut by several

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N-S granitic dikes too narrow to be shown in Figure 8. The deposit is partially and unconformably overlain by undisturbed, flatly-dipping Cambrian sediments.

Origin of the Ore.—Cumberlidge and Stone (6) believe that thermal metamorphism altered the mineralogy but did not alter the bulk chemistry or oxidation state of the primary iron formation to form the concentrating iron ore.

James (18, 19) and James *et al.* (21) believe that thermal metamorphism altered the mineralogy and changed the oxidation state, but that it did not change the bulk chemistry of the primary iron formation in forming the concentrating iron ore.

Van Hise and Leith (with Smyth) (34) believe that the iron silicates and carbonates in the primary iron formation were oxidized by meteoric solutions, and then the altered iron formation was metamorphosed to form the concentrating iron ore.

Paleomagnetic Results.—All the samples (D-1 to 15) were collected from well below the original surface in the mine open pit except for one sample (D-16) which was collected from a drift-stripped subcrop as shown in Figure 8. The samples of Vulcan iron formation were collected from all four members and from both sides of the ore body.

Fourteen samples (D-1 to 8 and 10 to 15) were collected from the Vulcan iron formation of which one sample (D-13) crumbled so that it could not be measured, and a second sample (D-7) from the E-W fault zone proved to have unstable remanence so that it was excluded. The four samples from the basal quartzite (D-1 and 3) and lower iron-bearing (D-2 and 15) members have fairly uniform magnetic characteristics which include stable remanence moments of low intensity (Table 2) which suggest a CRM. Their remanence directions form a cluster with precision parameters of 11.8 and 10.9 before and after unfolding, respectively (Fig. 7b). Either group may have statistical significance; however, the samples have similar tectonic corrections so that the fold test is not diagnostic. Eight samples came from the middle iron-bearing member (D-14) and its oxidized equivalent (D-8) and from the upper iron-bearing member (D-4, 5, 6, 11 and 12) and its hydrated equivalent (D-10). The single oxidized and hydrated samples do not appear to have any diagnostic magnetic property which warrents individual discussion and so they are included in the group of eight samples. These samples have a wide range of magnetic properties that reflects their mixed ferromagnetic mineralogy (as shown in Table 2). In particular, they have large unstable remanence moments due mostly to the presence of magnetite as shown in Figure 4e. Also, these samples have directions of maximum magnetic susceptibility that exceed the directions of minimum susceptibility by at least 40 percent in magnitude on the average. Unfolding causes the precision parameter for the remanence directions of those eight samples after AC demagnetization to decrease from 2.3 to 1.8 assuming the structure is a monocline and to 1.2 assuming a syncline. None of these values for the precision parameter has statistical significance, which suggests the presence of more than one type

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of remanence. However, the remanence inclinations have an average value of $10° \pm 9°$ (s.d.) relative to the bedding plane of the iron formation and randomly oriented declinations assuming a syncline as shown in Figure 7d whereas there is not a systematic pattern assuming a monocline. This remanence pattern indicates that the primary remanence directions of samples from the middle and upper iron-bearing members have been modified and thereby rotated into the bedding plane through metamorphism, and that the deposit has a synclinal structure. The co-existence of magnetite and hematite as shown in Figure 3b and noted by Cumberlidge and Stone (6), militates against any genetic hypotheses that advocate the alteration of the mineralogy by hydrothermal or meteoric solutions.

Four Cambrian samples were collected from sandstone (C-6) on the Menominee Range (Fig. 6), and from sandstone (D-16) and basal conglomerate (D-9A and 9B) at the Groveland Mine (Fig. 8). The two basal conglomerate samples yield divergent remanence directions as shown in Figure 7c. One sample (D-9B) has a specific gravity of 2.94 and a moderately stable remanence (Fig. 4f) which reflect the presence of the observed hematitic pebbles of iron formation. The other sample (D-9A) contains only a matrix material for it has a specific gravity of 2.68 which corresponds to values of 2.69 and 2.48 determined for the sandstone samples (C-6 and D-16, respectively). The latter three samples (C-6; D-9A and 16) have very uniform magnetic characteristics (Table 2) which include extremely stable remanence moments that are almost unchanged by AC demagnetization as shown in Figure 4c. The remanence directions of the three samples yield a post-folding precision parameter of 5.9 and the computed paleomagnetic pole position agrees closely with the other Cambrian positions (17). The remanence is probably a CRM associated with the ultra-fine hematite grains that surround the well-sorted quartz grains or oölites. The conglomerate test given by samples D-9A and D-9B is indicative of remanence stability since deposition of the conglomerate during the Cambrian period.

CONCLUSIONS

Animikie Iron Formations.—The remanence directions from these three examples of iron formation yield the same distinctive pattern in which the remanence directions are uniformly distributed about the bedding plane in inclination and randomly distributed in declination. Most geologists (20, 22, 23, 34) agree that the Negaunee iron formation of the Republic Mine, Marquette iron range, and the Vulcan iron formation of the Groveland Mine, Felch Mountain iron district, and of the Menominee iron range are contemporaneous sediments of Middle Animikie age. It is probable that they have somewhat similar geologic histories because they come from a small area in the same geologic province. Unmetamorphosed and slightly metamorphosed iron formations of the same age and region (31) and elsewhere (24, 35) have yielded well-grouped prefolding remanence directions from stable primary CRM. However, these three iron formations have been metamor-

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phosed in the Penokean orogeny (19) after burial by the Upper Animikie Michigamme series. If, during metamorphism, new ferromagnetic minerals were formed or if the Curie temperatures of the ferromagnetic minerals were exceeded, then the iron formations would have well-grouped post-folding CRM or TRM directions. This leads to the second general conclusion.

The primary remanence of these metamorphosed iron formations was to a large extent but not entirely modified during metamorphism. Most likely the iron formations were deposited during the Middle Animikie as chemical sediments of the hematite-banded and magnetite-banded oxide facies with a primary CRM. Several bits of evidence indicate that a residual of this primary component is still present. First, the prefolding remanence direction (74°, 41°; $\alpha 63 - 14°$) found for the basal and lower iron-bearing members of the Vulcan iron formation at the Groveland Mine after tectonic correction is similar in inclination, though not in declination due to relative tectonic rotation, to that (169°, 19°; $\alpha 63 - 20°$) found for the primary Negaunee iron formation at the east end of the Marquette Range (32). Second, if the conglomeratic iron formation at the Republic Mine has a sedimentary origin, then the divergence of the remanence directions within and between its samples suggests that some of the primary remanence component has survived metamorphism. Third, the co-existence of hematite and magnetite in both the Republic and Groveland Mine deposits (Figs. 3a, 3b; reference 5), the superposition of the magnetite-banded oxide facies over the hematite-banded oxide facies at the Groveland Mine, and the preservation of unaltered and undeformed primary structures and textures in the iron formations (21), all suggest that the iron formations were not altered by groundwater or hydrothermal solutions before burial, during diagenesis, or later. Thus it seems probable that the primary remanence was significantly modified in some way during metamorphism to yield the distinctive pattern but not completely eliminated.

The process of remanence modification is not clear but it seems likely that metamorphism caused a dominant stable PPRM, PTRM or PCRM (partial pressure, thermal, or chemical remanent magnetization, respectively) component to be superimposed on the primary component (17). The process is obscured by the complexity of the geology which involves the interrelation of mixed ferromagnetic mineralogy with a complex stress and thermal history. Such a combination of components could yield the required remanence pattern. James *et al.* (19, 21) consider that the metamorphism was dominantly thermal because they found that: (a) the primary oölites in the iron formation of the Groveland Mine are undeformed by stress; (b) the crystal dimensions of the minerals in the iron formation increase steadily with increased thermal metamorphic grade; and (c) new mineral facies are progressively developed in the iron formation to delimit the thermal metamorphic grades. Also, they believe that the increased grain size made economic beneficiation of the iron formation possible. Such a transformation in the ferromagnetic minerals would result in a PTRM or PCRM. The author considers that dynamic metamorphism was very important because:

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(a) the remanence pattern of these three metamorphosed iron formations is independent of their thermal metamorphic grade, which reached the chlorite-biotite facies on the Menominee Range, the staurolite facies at the Groveland Mine, and the sillimanite facies at the Republic Mine which represent low-, medium-, and high-grade intensities, respectively (19); (b) all three iron formations have been strongly deformed to about the same extent by dynamic metamorphism, which caused them to be tightly folded into synclines at the Republic and Groveland Mines and into synclines and anticlines on the Menominee Range; (c) the finding of the remanence pattern is dependent on the correct structural interpretation which, incidentally, supports the synclinal structural interpretation for the Groveland Mine where the structural interpretation is in some doubt; (d) the prominent magnetic anisotropy illustrated by the magnetic susceptibility of the iron formations in which the average ratio of the maximum axis to minimum axis is about 1.5 : 1.0 with the maximum axis closely parallel to the bedding plane, has a much greater magnitude than found in unmetamorphosed iron formations (30) and is probably a response to increased pressure; (e) extreme dynamic metamorphism in similar Precambrian iron formation in the USSR has strongly affected the remanence and caused it to become unstable (36); and (f) as Nagata (25) has shown, even the magnetic properties of aggregations of simple ferromagnetic minerals behave unpredictably in stress fields. This evidence suggests that a significant PPRM component is present in the remanence. Also, a combination of primary and secondary components suggests the explanation of why the remanence of the middle and upper iron-bearing members of the Vulcan iron formation at the Groveland Mine has been modified whereas that of basal and lower members has not been modified by metamorphism. The former contain mostly coarse-grained magnetite that is much more susceptible to dynamic and thermal remanence alteration than the latter, which contain mostly fine-grained hematite. This interpretation is supported by the evidence of the two samples (D-6 and 10) from the upper iron-bearing members that have hematite as a major constituent and which give remanence directions that are close to those of the basal and lower iron-bearing members (Figs. 7b, 7d). It is noted that the remanence of the middle and upper members may have been more affected than that of the basal and lower members by dynamic metamorphism because of the former's position in the center of the fold structure. The direction of the secondary component would depend on the relative orientations of the sample, the adjacent rocks and the Earth's magnetic field in addition to temperature, pressure, and chemical considerations. The bias of the remanence directions about the bedding plane could possibly be explained by the "infinite magnetic sheet" hypothesis of Strangway (28). Summarizing, there does not appear to be a thoroughly satisfactory explanation of how the primary CRM of the iron formations was modified during metamorphism to form the distinctive pattern although a PPRM component appears to be the most attractive explanation.

Keweenawan Diorites.—The Keweenawan diorites have a post-folding TRM with a Lower Keweenawan direction. After the Penokean orogeny

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and metamorphism, the Lower Keweenawan diorites were intruded into the folded iron formations as sills and dikes as shown at the Republic Mine, to yield a post-folding TRM with a Lower Keweenawan direction. The ferromagnetic minerals in the iron formation adjacent to the diorites were heated through their Curie point temperatures and perhaps recrystallized as different minerals so that they now have a post-folding TRM or CRM of Lower Keweenawan age and belong to the magnetite-banded oxide facies. During the remainder of the Keweenawan period, the Grenville orogeny, and the infra-Cambrian period, the iron formation and diorites were not altered or subjected to significant structural rotation or deformation.

Cambrian Sediments.—The Cambrian sediments have a very stable primary CRM or DRM in a Cambrian direction. The Cambrian sandstones and conglomerates were deposited unconformably over the Precambrian formations with a stable primary CRM or DRM. They, and the older formations, have been essentially unchanged or unmoved since the Cambrian.

Summary.—The paleomagnetic results are consistent with the geologic evidence. They support the hypothesis that the Negaunee and Vulcan iron formations of the Republic Mine, Groveland Mine, and Menominee Range were deposited as iron formations of the magnetite-banded and hematite-banded oxide facies during the Middle Animikie. During the Upper Animikie they were buried by the Michigamme series, and then during the Penokean orogeny they were dynamically and thermally metamorphosed. These metamorphic events probably did not change appreciably the bulk chemistry, the oxidation state, or the ferromagnetic mineralogy of the iron formations although they did alter the texture and the magnetic domain structure of the ferromagnetic minerals through dominantly stress effects. Except for a narrow zone of contact thermal metamorphism associated with the later intrusion of the Keweenawan diorites, the iron formations have not been subsequently altered. This interpretation of the genesis of these metamorphosed iron formations that form the concentrating ores of the Republic and Groveland Mines agrees most closely with that proposed by Cumberlidge and Stone (6) for the Groveland Mine.

Only the paleomagnetic pole positions derived from the Keweenawan diorite and Cambrian sandstone are thought to be reasonably reliable, and they are given in Table 2. The Animikie strata are thought to have been rotated so that a tectonic correction cannot be applied to derive a reliable pole position.

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